

RECONNAISSANCE & NETWORK ENUMERATION

Service Enumeration Report

Target: 192.168.1.3 (Metasploitable 2)

Scanner: Kali Linux 192.168.1.4

Enumeration Method

Service enumeration was performed using Nmap version detection and Nmap default scripts. The commands used were:

```
nmap -sV 192.168.1.3
```

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nmap -sC -sV 192.168.1.3
```

The -sV option attempts to identify the service and software version running on each open port. The -sC option runs Nmap's default NSE scripts, which can provide additional information about selected services and host configuration.

Service Enumeration Results

Command: nmap -sV 192.168.1.3

Port	Service	Detected software/version
21/tcp	ftp	vsftpd 2.3.4
22/tcp	Ssh	OpenSSH 4.7p1 Debian 8ubuntu1 (protocol 2.0)
23/tcp	telnet	Linux telnetd
25/tcp	Smtp	Postfix smtpd
53/tcp	Domain	ISC BIND 9.4.2
80/tcp	http	Apache httpd 2.2.8 ((Ubuntu) DAV/2)
111/tcp	Rpcbind	2 (RPC #100000)

139/tcp	netbios-ssn	Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
445/tcp	netbios-ssn	Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
512/tcp	Exec	netkit-rsh rexecd
513/tcp	Login	OpenBSD or Solaris rlogind
514/tcp	Tcpwrapped	tcpwrapped
1099/tcp	java-rmi	GNU Classpath grmiregistry
1524/tcp	Bindshell	Metasploitable root shell
2049/tcp	Nfs	2-4 (RPC #100003)
2121/tcp	ftp	ProFTPD 1.3.1
3306/tcp	Mysql	MySQL 5.0.51a-3ubuntu5
5432/tcp	Postgresql	PostgreSQL DB 8.3.0-8.3.7
5900/tcp	Vnc	VNC (protocol 3.3)
6000/tcp	X11	(access denied)
6667/tcp	Irc	UnrealIRCd
8009/tcp	ajp13	Apache Jserv (Protocol v1.3)
8180/tcp	http	Apache Tomcat/Coyote JSP engine 1.1

Service Analysis

FTP (21/tcp): Nmap identified vsftpd 2.3.4 and reported that anonymous FTP login was allowed. The scan also reported plain-text FTP control and data connections. Access controls and the need for FTP should be reviewed.

SSH (22/tcp): OpenSSH 4.7p1 was identified. An older implementation that should be reviewed and appropriately restricted.

Telnet (23/tcp): Legacy remote terminal service. Because Telnet does not provide the same protection as modern encrypted remote administration, unnecessary use should be avoided.

SMTP (25/tcp): Postfix was identified. Nmap reported SMTP capabilities including VRFY, ETRN, and STARTTLS, as well as SSLv2 support.

DNS (53/tcp): ISC BIND 9.4.2 was detected. DNS exposure should be reviewed for secure configuration.

HTTP (80/tcp): Apache 2.2.8 on Ubuntu was identified, with the title 'Metasploitable2 - Linux'.

SMB (139/445): Samba was detected. The detailed scan identified Samba 3.0.20-Debian on port 445 and reported SMB message signing as disabled.

RSH/Rlogin (512/513/514): Legacy remote access services were detected. These should be disabled when not required.

NFS (2049): Network File System was detected. Exported resources and access restrictions should be reviewed.

Databases (3306/5432): MySQL and PostgreSQL were reachable. Database services should normally be restricted to authorized networks.

VNC (5900): VNC protocol 3.3 was identified. Remote desktop exposure should be tightly controlled.

IRC (6667): UnrealIRCd was detected. The service reported the host name irc.Metasploitable.LAN.

Java RMI/AJP/Tomcat (1099/8009/8180): Java RMI, Apache Jserv/AJP13, and Apache Tomcat were identified. These services add application-server functionality and should be restricted.

Additional Script Findings

Command: nmap -sC -sV 192.168.1.3

Anonymous FTP: Nmap reported anonymous FTP login allowed on port 21.

SMTP: SSLv2 was reported as supported, and the certificate shown was from 2010.

SMB: Nmap reported SMB message signing disabled and identified the OS as Unix (Samba 3.0.20-Debian).

MySQL: The mysql-info script returned protocol and capability information for MySQL 5.0.51a.

Web services: Apache on port 80 and Tomcat on port 8180 were identified. **Host information:** Nmap identified the system as Unix/Linux and reported the FQDN metasploitable.localdomain.

Potential Security Concerns

Service	Observed concern	Priority for review
FTP 21	Anonymous access and plain-text FTP	High
Telnet 23	Legacy remote access	High
SMB 139/445	SMB exposed; signing reported disabled	High

HTTP 80	Older Apache version exposed	High
RSH/Rlogin 512-514	Legacy remote-access services exposed	High
Databases 3306/5432	Database services network accessible	High
VNC 5900	Remote graphical access exposed	Medium-High
Tomcat/AJP 8009/8180	Application-server services exposed	Medium-High
DNS 53	Older BIND version identified	Medium
SSH 22	Older OpenSSH version identified	Medium

Recommended Security Actions

- Disable services that are not required.
- Replace legacy remote-access protocols such as Telnet and RSH/Rlogin with secure alternatives.
- Apply appropriate security updates to exposed software.
- Restrict database, SMB, VNC and application-server services to authorized networks.
- Review FTP anonymous access and avoid plain-text protocols where possible.
- Apply secure SMB configuration, including appropriate message-signing settings where required.
- Use network segmentation and firewall rules to reduce unnecessary exposure.
- Maintain an accurate inventory of exposed services and review it regularly.

Conclusion

Service enumeration of 192.168.1.3 identified multiple open TCP services, including remote-access services, file-sharing services, web services, databases,

and other network applications. Nmap version detection and default scripts provided useful information about software versions and selected configurations.